

Aadarsh Upadhyay

Software Engineer / Architect

CURRICULUM VITAE

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India · remote-first · open to relocate · outside-India visa sponsorship welcome

I am a React-first full-stack engineer and product architect. My home stack is **CSS, React, TypeScript, Next.js, Electron, Node**, and production deploy. Most of my product depth comes from Appson work on **ComplyV** (formerly Compliance Sarathi) and **Wordibly**, where I owned governed AI writes, multi-tenant access, deadline math, and production UI. Public proof is **searchts** on PyPI through **0.13.0** (I push, review, and release). I also own [Grove](#), **Dooper**, **GlyphMaps**, and **capad.fyi**; Hare is not a SaaS yet; I shipped **Hare Bot**. I use AI assistants to ship and review hard, with custom skills and runbooks so agent desks follow product constraints. I am finishing BCA (Honours) at The Maharaja Sayajirao University of Baroda, Faculty of Science, Dept. of Computer Applications (expected 2028). Looking for remote or India roles that hire for ownership of shipped systems. Open to relocate; outside-India roles with visa sponsorship welcome.

EXPERIENCE

Software Engineer and Architect, Appson Technologies (ComplyV, formerly Compliance Sarathi)

Jan 2026 - 31 Jul 2026

Indian **corporate-governance SaaS**: MCA/ROC company data, statutory deadlines and penalties, resolution and minutes drafting, AI chat (Vee) with governed writes. Dual UI (legacy CRA + /new-ui). **Primary engineer and architect** on a small team (~553/647 commits (~84%)). Not sole developer. Last commit and written handover 31 Jul 2026. Stack: **React 18** (CRA), React Router, Tailwind, Bootstrap, **Node/Express**, **MongoDB/Mongoose** (34 models), **JWT/RBAC**, WebSocket, OpenAI/Gemini/Anthropic, AWS S3, Twilio, Brevo/Resend, Word/Excel/PDF.

Architecture notes (from shipped systems): Vee never mutates data directly. Writes go through **propose-then-confirm** confirmation cards, then a separate authenticated endpoint that re-checks auth, company access, and business rules with **idempotency keys** and **audit logs**. A **three-tier truth wall** keeps due dates and penalties on the **deadline engine** only; statutory drafts reuse the product drafting pipeline; eval cases bait **number leakage** on conceptual answers. Company access is an **ownership-and-grant graph** over CIN (not a naive tenantId); LLM tools receive a CIN string resolved inside **assertCompanyAccess**. The **SHA-256 append-only audit log** uses immutability hooks, deterministic hashing, a write mutex, and an **admin integrity verifier** (tamper-resistant, not tamper-proof). The **AGM/deadline engine** is ~1,400 LOC of pure computation for multiple MCA forms with entity-aware deadlines and per-day penalties; the draft linter is code, not model self-grading.

- Owned the **React and Node product surface**: auth, multi-tenant company access, deadline math, minutes/docs drafting, Vee agentic chat. Commit share ~553/647 (~84%).
- Built **propose-then-confirm writes**. The model never mutates data. Proposals show as confirmation cards. A separate authenticated endpoint re-checks auth, company access, and business rules with **idempotency keys** and **audit logs**.
- Built a **three-tier truth wall**. Due dates and penalties come only from the **deadline engine**. Statutory drafts reuse the product drafting pipeline. Eval cases bait **number leakage** on conceptual answers.
- Implemented a **SHA-256 append-only audit log** with immutability hooks, deterministic hashing, a write mutex, and an **admin integrity verifier**. Tamper-resistant, not tamper-proof.
- Multi-tenancy** without a naive tenantId. Company access is an **ownership-and-grant graph** over CIN. LLM tools get a CIN string that resolves inside **assertCompanyAccess**.
- Maintained the **AGM/deadline engine** (~1,400 LOC of pure computation) for multiple MCA forms with entity-aware deadlines and per-day penalties. Draft linter is code, not model self-grading. Fixed **LLP fee slabs** and a **DIR-3 KYC triennial rule** that would have been wrong from 2027.
- Registered **34 Mongoose models** across identity, tenancy, documents, AI runtime, governance, and commerce. Backend tests: ~40 suites / ~2,852 tests (technical sheet).
- Found **auth gaps** while designing agent tools (**deny-by-default** draft visibility and writes). Made **idempotency a unique index in the database**, not an in-memory lock.
- Wrote the 31 Jul 2026 handover: architecture, known issues, secret rotation.

Junior Software Engineer, Appson Technologies (Wordibly)

Jun 2025 - early 2026

US **transcription, translation, and captioning platform** (human, hybrid, AI tiers). Strongest personal ownership: **tops-transcript-editor (Vite/React)**. Broader platform (tops-api, tops-new-ui, tops-old-ui) is large-team CodeCommit with thinner personal share. Stack exposure: React, Node, Python, MongoDB, AWS; Zoho, QuickBooks, Zapier, Otter.

- Owned **tops-transcript-editor (~107-113 commits)**: **Vite/React transcript editor** with find/replace and related editing flows for production transcription work.
- Contributed on the wider **CodeCommit platform** (tops-api, tops-new-ui, tops-old-ui) with a thinner personal share: **order/upload surfaces, workforce dashboards, and editor workflows** for PMs, transcribers, and QA.
- Daily Scrum with Appson and Wordibly. **Sprint delivery** with **no visual regressions** on production-bound UI.

AI Training Engineer, Turing (via Appson)

~2 weeks, 2025

Meta Llama training data and evaluation.

OPEN SOURCE AND PRODUCTS

searchts

Stack: Python, MIT · PyPI v0.13.0 · CLI + MCP · github.com/capad-xyz/searchts

Role: My product. I own the unlocker, **search fusion**, sanitizer, MCP tools, share-link extractors, and **releases** (push, review, ship through 0.13.0).

Outcomes: **Keyless web unlocker** plus read/search/transcribe/grab for AI agents. Escalates browser-fingerprinted fetch → JS-render relay → stealth browser, then Markdown extraction. Defaults refuse paid residential proxies and keyed commercial unlockers so personal-volume use stays free. Fail-loud on login walls and CAPTCHAs. The local MCP/connector for searchts is built; the remote connector will be built soon. Agent-facing docs/skill pack (AGENTS.md, RUNBOOK, CLAUDE.md, SKILL.md) so assistants use searchts the intended way, not as a raw fetch wrapper.

Hare

Stack / status: not a SaaS yet · GitHub App searchts-hare[bot] · @hare · R1c · dogfooded on searchts

Role: I design the full **PR review surface**: structured summary, severity findings, inline bubbles, and replies.

Outcomes: Not a SaaS yet: a **review-and-report GitHub App** I dogfood on searchts. Grounded in **CI and diff evidence**, non-blocking feedback without a SaaS lock-in. Trigger: PR open, push, or /hare → gather → **one model JSON** → posted review (R1c). Writer model never Hare model. **Review/report only**; never blocking APPROVE/REQUEST_CHANGES. Not a SaaS yet; no CodeRabbit parity claims.

Hare Bot

Stack: Cursor Grok Bot · PR review/report

Role: I design review loops as a Cursor Grok Bot for maintainers and agent desks.

Outcomes: Reads **PR body/diff/commits/CI as evidence**, can checkout and run tests, then posts a **COMMENT review** with summary, severity findings, inline bubbles, replies, and Models attribution: more of the review loop than a single dump.

Review/report only. Hare Bot does not fix, push, or merge unless asked; never blocking APPROVE/REQUEST_CHANGES. Chat-side template; Hare App is a separate surface.

capad.fyi

Stack: Next.js, React, Sanity, Cloudflare Workers / OpenNext · capad.fyi

Role: My site and product desk.

Outcomes: **Liquid-glass UI** (CSS backdrop-filter + SVG refraction), **Sanity-backed projects / timeline / testimonials / resume** with HMAC webhook revalidation into KV-backed ISR, multi-host Worker for capad.fyi and glyphmaps.capad.fyi, adaptive perf-lite path, **GitHub Actions deploy**.

MSU Halls Register

Stack / status: design prototype · msu.capad.fyi

Role: Design and static prototype for institutional alumni register.

Outcomes: Alumni-register prototype for the Halls of Residence at The Maharaja Sayajirao University of Baroda (**16 halls**).

Eight-page static design system revised after Office of the Chief Warden feedback. Live prototype; engagement paused, not under active contract.

GlyphMaps

Stack: Kotlin / Android, v1.0.0 · github.com/capad-xyz/GlyphMaps

Role: Solo Android product for Nothing Glyph Matrix navigation.

Outcomes: **Google Maps Live Updates** on the **Nothing Glyph Matrix**, built for Nothing Phone (4a) Pro; works on Phone 3 (needs work). **Next-turn arrows** on the phone back, **no Maps API key**. Skipped the throttled Glyph Toy redraw path. Drives the matrix while navigating, then releases it so other Glyph toys return.

Dooper

Stack: repo beep-beep-oss · Tauri 2 + React/TypeScript + Matrix · Alpha Phase 1

Role: Product owner / builder of AGPL self-hostable universal inbox.

Outcomes: **AGPL self-hostable universal inbox:** Synapse, mautrix bridges, Tauri 2 + React. Messaging sync without paywalled delay. **Phase 1 verified on real bridged WhatsApp** (login, inbox, history, **optimistic send**, **session persistence**). Product name Dooper; repo is beep-beep-oss.

Grove

Stack: Electron + React + TypeScript, GPL-3.0 · alpha · github.com/capad-xyz/grove

Role: Rebuilt and own Grove as worktree-aware git review beside AI editors.

Outcomes: Rebuilt on Electron, React, and TypeScript with a **headless Node git engine**. **Worktree-aware git review:** lane-drawn commit graph, live refresh, **find-in-diff**, **local-CLI commit messages**. No paid in-house model lock-in.

burncard

Stack: local CLI · npx burncard

Role / outcomes: Local Claude Code / Codex usage telemetry.

SELECTED OPEN-SOURCE CONTRIBUTIONS

wmux

Contributor · [amirlehman/wmux](https://github.com/amirlehman/wmux) · not my project

I dogfood wmx in daily agent work. I file issues when I hit bugs; I land PRs when I have a fix. As of 2026-10-02: **8 closed issues**, **4 merged PRs** ([#135](#), [#138](#), [#153](#), [#258](#)) on **diff-pane freezes**, **CLI timeouts**, and agent-browser install discovery. Best proof: [#135](#): root-caused a **main-process freeze** when a diff pane watched a non-git cwd; landed **bounded snapshot walks**.

SELECTED SYSTEMS

ComplyV governed-write path. Propose-then-confirm cards, separate authenticated confirm endpoint, idempotency as a unique database index, deny-by-default draft visibility, and SHA-256 append-only audit with admin integrity verifier. Tamper-resistant, not tamper-proof.

ComplyV truth wall and deadline engine. Due dates and penalties only from the ~1,400 LOC deadline engine; statutory drafts from the product drafting pipeline; eval cases bait number leakage. Multi-tenancy via ownership-and-grant graph over CIN resolved in assertCompanyAccess.

searchts unlocker and MCP. Keyless escalation (fingerprinted fetch → JS-render relay → stealth browser) with Markdown extraction; local MCP/connector built, remote coming soon; releases through PyPI 0.13.0 with agent skill pack.

capad.fyi content desk. Sanity-backed projects/timeline/testimonials/resume, HMAC webhook revalidation into KV-backed ISR, multi-host Cloudflare Worker (capad.fyi + glyphmaps.capad.fyi), liquid-glass UI, GitHub Actions deploy.

EDUCATION

Bachelor of Computer Applications (Honours)

Expected 2028

The Maharaja Sayajirao University of Baroda, Faculty of Science, Department of Computer Applications. Ongoing undergraduate programme with expected completion in 2028.

Product Space PM Fellowship, Top Fellow

2025 - 2026

Product management fellowship; recognized as Top Fellow for the 2025-2026 cohort.

SKILLS

Tech stack	React , React Three Fiber (R3F), CSS , design systems, Tailwind, Bootstrap, Framer Motion, React Router, Next.js (App Router), Vite, Electron . Comfortable from layout through production deploy .
Design / UI-UX	UI/UX , Figma , design systems , product UI from layout through ship. AI-assisted design drafts with Claude Design and Stitch. Proof: ComplyV /new-ui , Wordibly transcript editor, capad.fyi liquid-glass, MSU Halls register prototype, GlyphMaps matrix UI.
Full-stack	TypeScript , JavaScript, Node.js , Express, MongoDB/Mongoose , REST/JWT , Python (CLI), Git, GitHub Actions, MCP , Cloudflare Workers / OpenNext , Sanity. End-to-end ownership of APIs, auth, and data models on products I ship.
AI / tools	Grok / Grok Bots (incl. Hare Bot), Claude ; daily drivers Oh My Pi (OMP) and GrokCLI; MCP , T3 Connect, Cowork, Custom Connectors, WebMCP; local searchts MCP/connector built (remote soon). Custom skills / AGENTS.md / RUNBOOK; writer ≠ Hare reviewer; PR text/diffs/CI as evidence , not instructions.
AI shipping	I own push and review on searchts and other repos: product decisions , release notes , merge-ready diffs , documented refusals of paid unlocker defaults.
Agent skills	Custom agent skills , runbooks, AGENTS.md , and RUNBOOK so multi-agent desks follow product constraints instead of improvising.
Systems	Agent tool design , authz , multi-tenancy , audit logs , eval harnesses , prompt-vs-code safety boundaries . Applied on ComplyV (truth wall, governed writes) and searchts (unlocker defaults, MCP).
Ship / DevOps	GitHub Actions (CI, release, Hare review on searchts), Cloudflare Workers / OpenNext (capad.fyi), Docker, Tailscale, AWS S3. Comfortable operating what I ship; not applying as a dedicated DevOps engineer yet.
Also	Tauri , Sanity CMS , Twilio, Brevo/Resend; Rust/Kotlin with AI assist (Dooper, GlyphMaps). Not targeting Rust/Kotlin/Go-primary roles.